

ETS052

Lecture 5

2006

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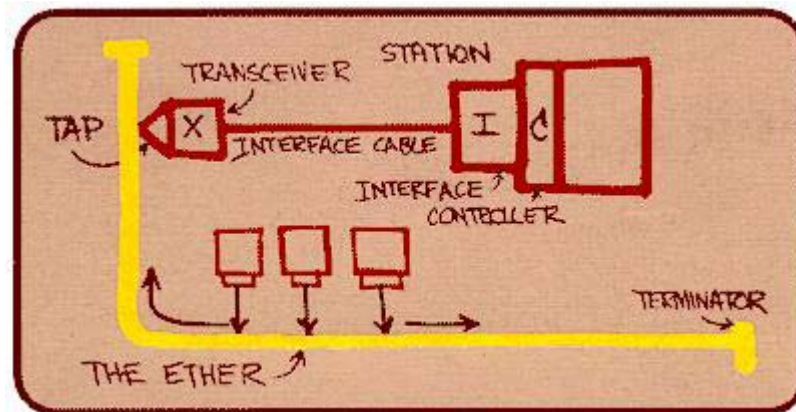
Content

Link technologies

- ◆ Ethernet
- ◆ VLAN
- ◆ WLAN
- ◆ Bluetooth

Standard Ethernet

- ◆ Invented at Xerox Palo Alto by Bob Metcalfe
- ◆ 1973
- ◆ Developed over time
- ◆ The Ethernet II version is used
- ◆ Can coexist with IEEE 802.3 Ethernet



IEEE Project 802

- ◆ Started in 1985

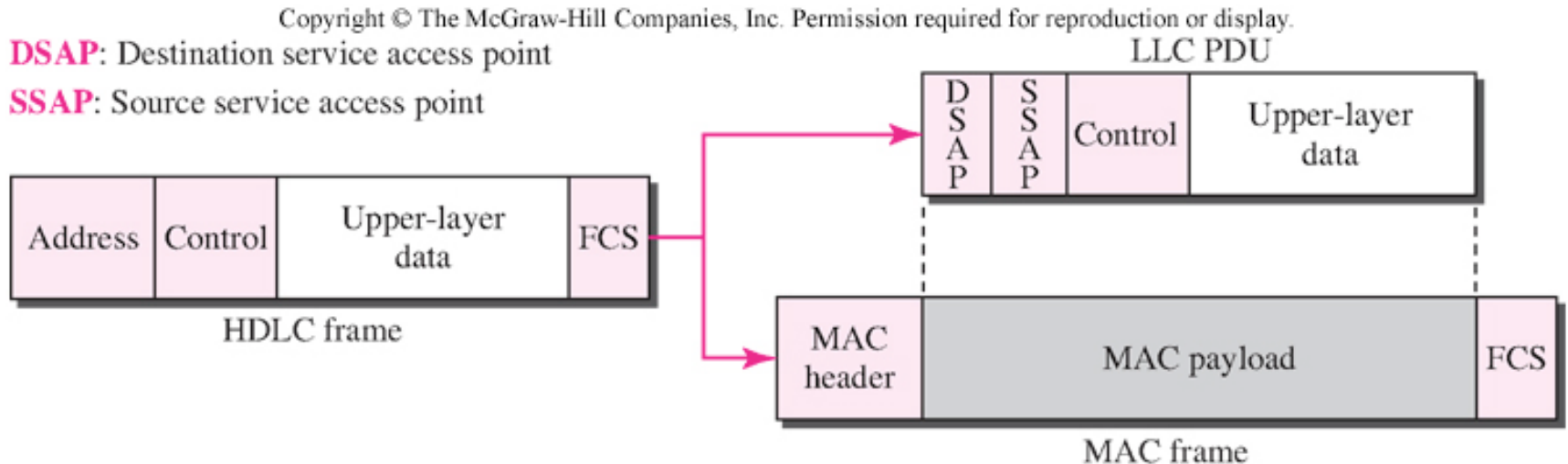
- ◆ Goal:

*to set standards to enable intercommunication
among equipment from a variety of
manufacturers*

- ◆ By this time Ethernet and TCP/IP already existed

IEEE 802 major characteristics

- ◆ Data link layer divided
 - Logical Link Control (LLC)
 - Medium Access Control (MAC)



IEEE 802.2 LLC

- ◆ Flow control
- ◆ Error control
- ◆ Part of link layer framing
 - Service Access Points
 - Control
- ◆ Only Service Access Point equivalent found in Ethernet = Type Field

IEEE 802.3 MAC & Ethernet

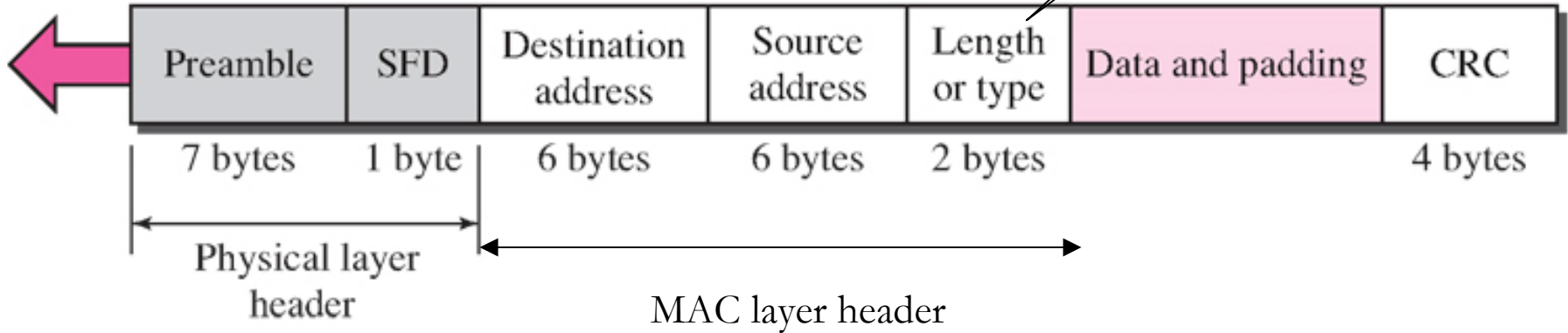
- ◆ Access Method: CSMA/CD
 - Carrier Sense
 - Multiple Access
 - Collision Detection
- ◆ Part of link layer framing (802.3)
 - Fields
 - Frame length

IEEE 802.3 and Ethernet MAC & Physical Headers

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Preamble: 56 bits of alternating 1s and 0s.

SFD: Start frame delimiter, flag (10101011)

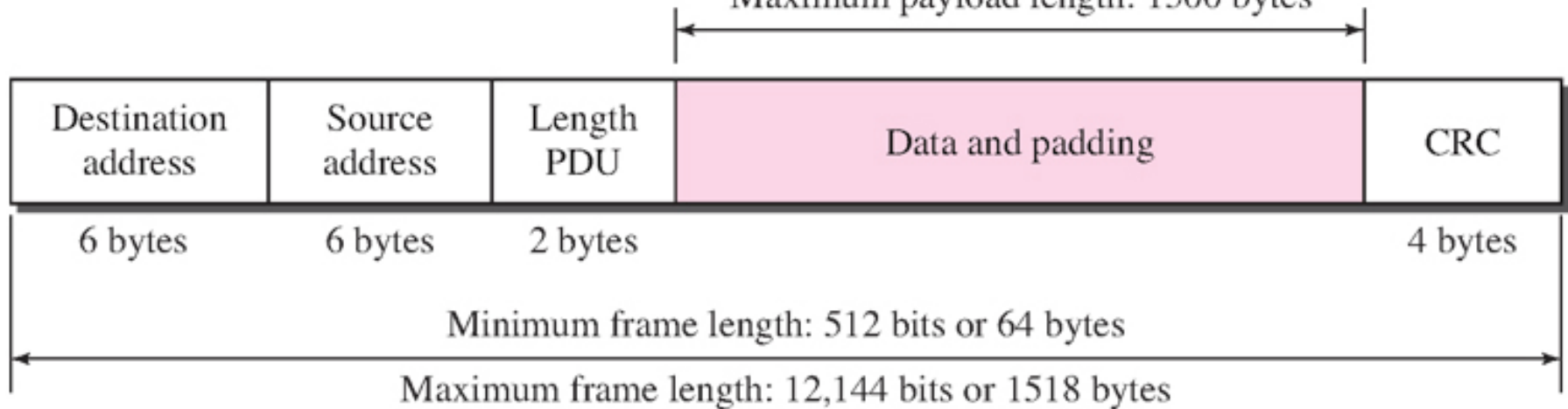


IEEE 802.3 and Ethernet Frame Length

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Minimum payload length: 46 bytes

Maximum payload length: 1500 bytes



IEEE 802.3 & Ethernet Phys. Layer

- ◆ Thick coax = “frozen garden hose”
 - ◆ Thin coax
 - ◆ Twisted Pair
 - ◆ Fibre
-
- ◆ Max capacity 10Mbps
 - Hard to exceed 7 Mbps in practical use

Increase Capacity

- ◆ Increase transfer speed
 - 100 Mbps, 802.3u
 - 1 Gbps, 802.3z
 - 10 Gbps, 802.3ae
 - Higher Speed Study Group, 802.3 HSSG
- ◆ Minimize risk for collisions
 - Decrease collision domains with bridges
 - Full duplex where possible

VLAN (network equipment)

Collision domains sharing physical resources

- ◆ In active network equipment: Tables
 - Complete Link Layer address vs physical port per VLAN
 - Separation of frames (hosts)
 - Per connection port
 - Per MAC-address
 - Per upper layer address (!)

VLAN (links)

Collision domains sharing physical resources

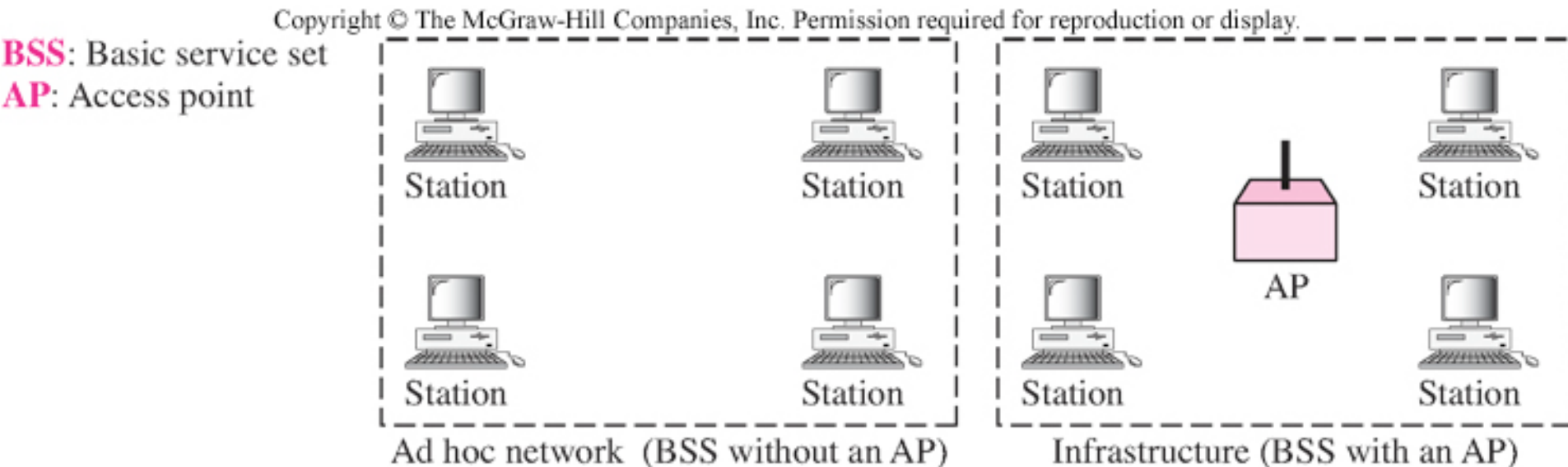
- ◆ On Links; also called Trunks

- Address tables
- Tagging
 - 802.1q adds control fields in non used fields
- Framing
 - Cisco ISL adds control fields (encapsulation)
 - Non standard ethernet frames (!)
- Time Division Multiplex
 - One time slot per VLAN

WLAN

Wireless LAN

- ◆ Radio based or Infrared
- ◆ Two modes

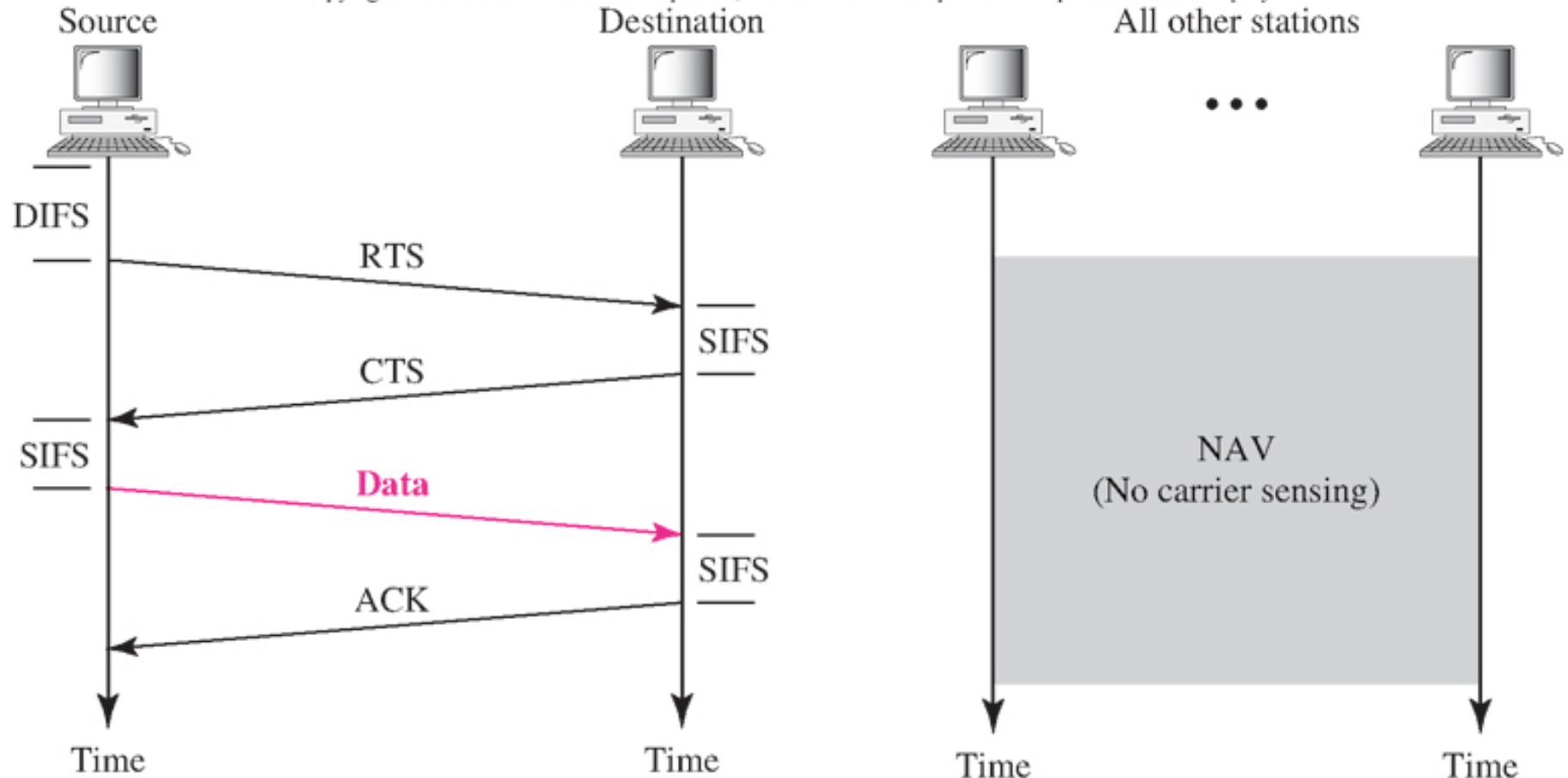


IEEE 802.11

- ◆ LLC = IEEE 802.2
- ◆ MAC divided
 - (PCF, DCF; not covered in course)
- ◆ MAC Access Method = CSMA/CA
 - Listen if media is free
 - Avoid collisions

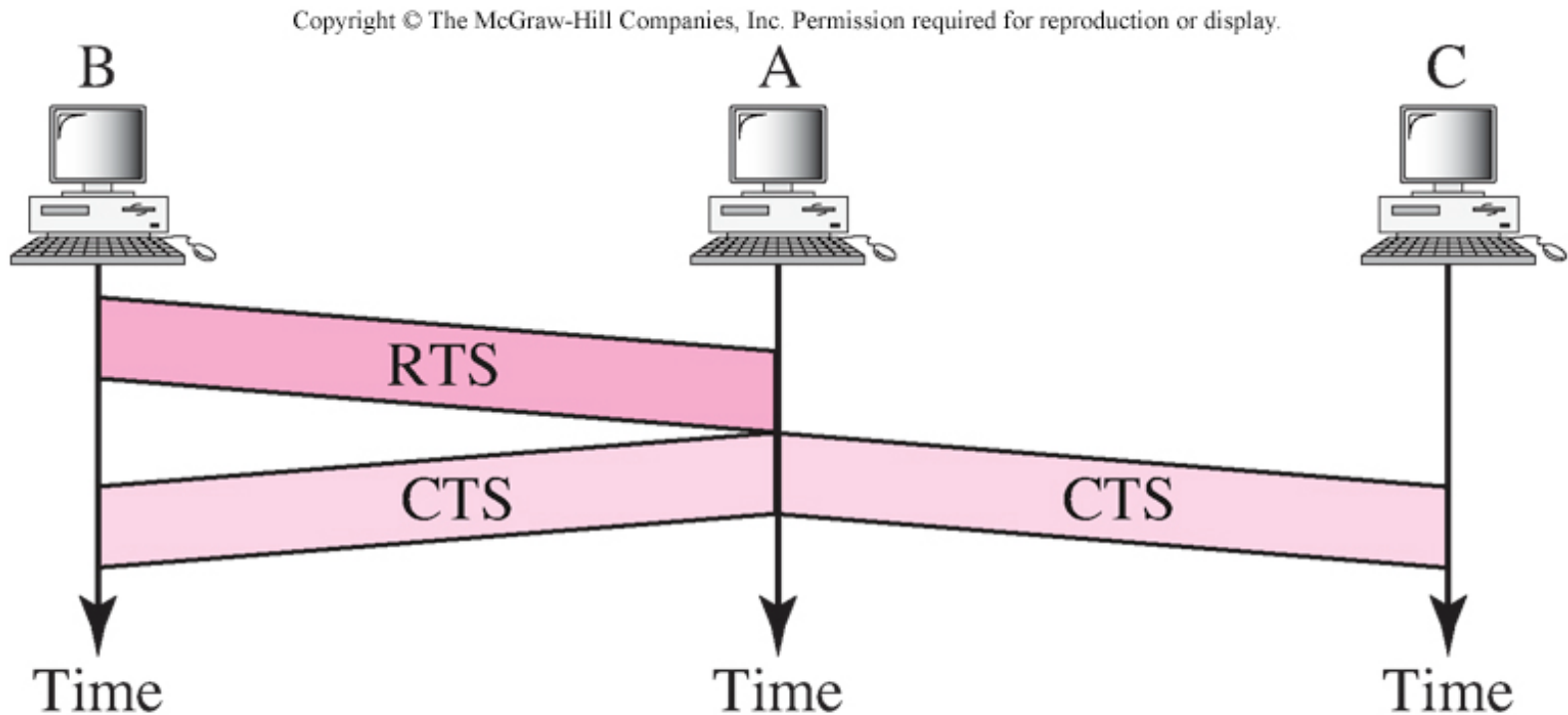
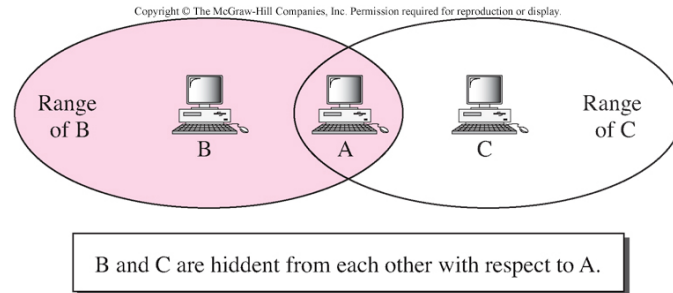
IEEE 802.11 Collision Avoidance

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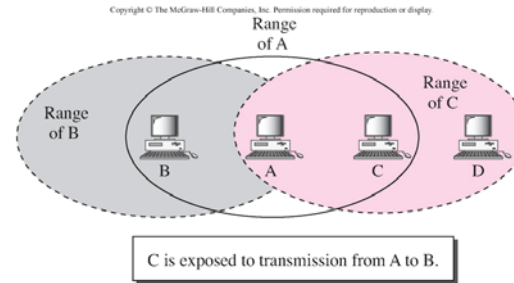
Collision Avoidance Problem

Hidden Node



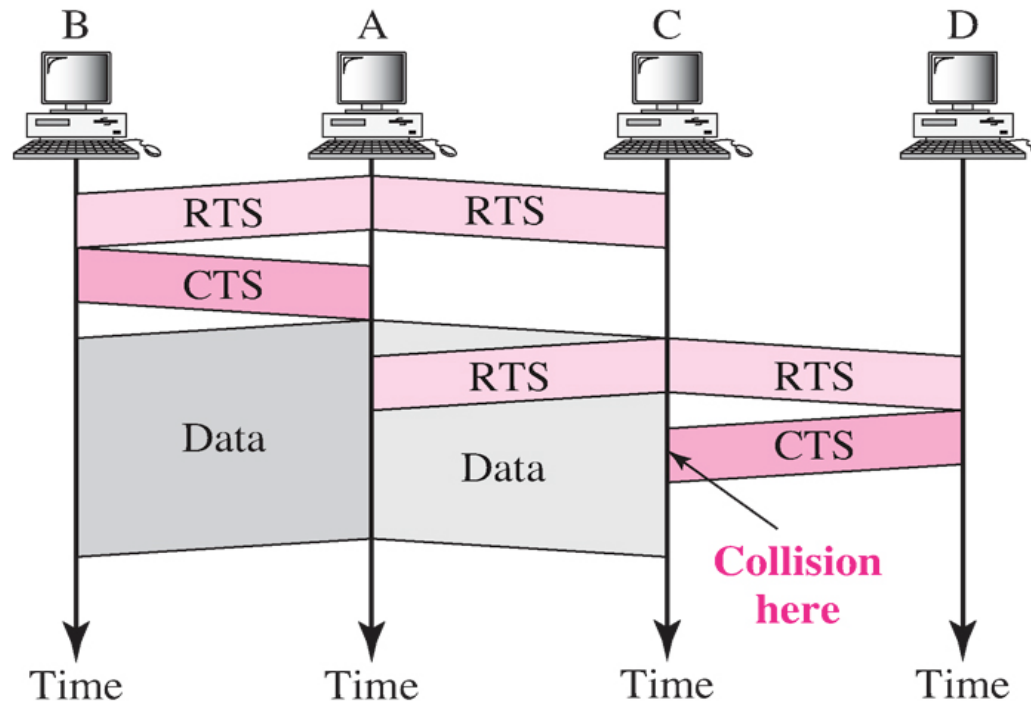
Collision Avoidance Problem

Exposed Node



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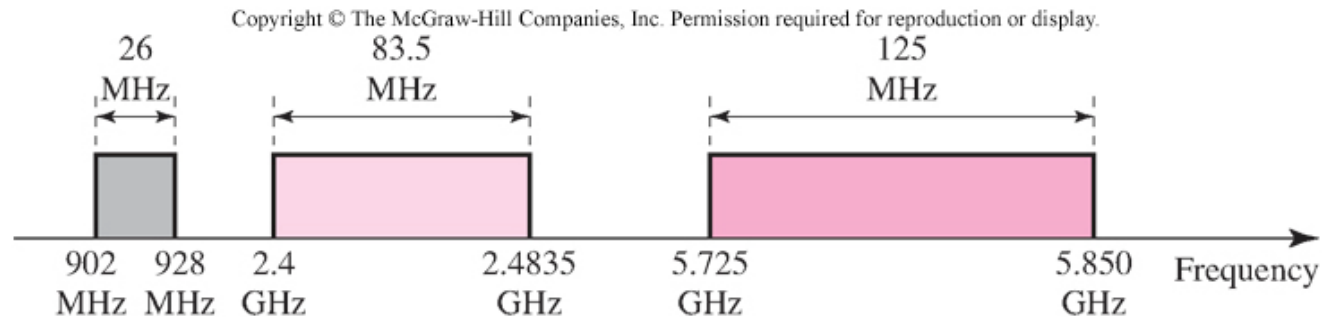
**Exposed to
A's transmission**



IEEE 802.11 Physical

◆ Radio based

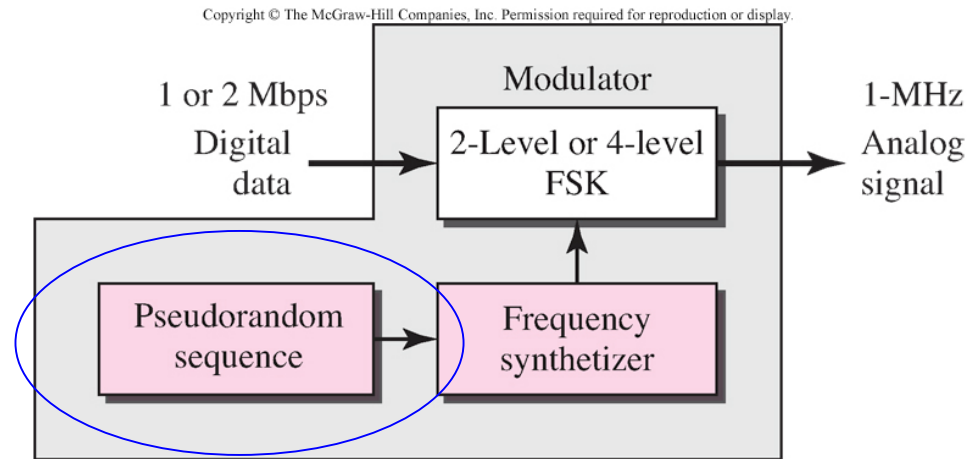
- FHSS (Frequency Hop Spread Spectrum)
- DSSS (Direct Sequence Spread Spectrum)



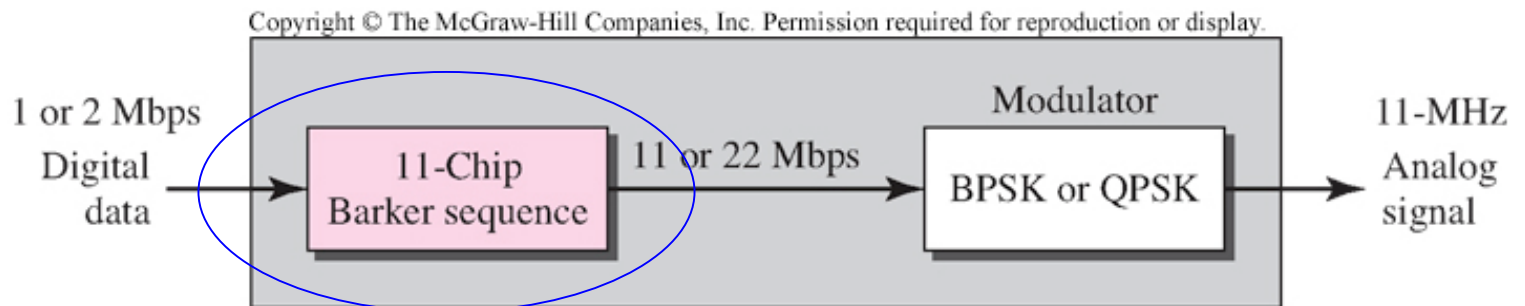
◆ Infrared

Spread Spectrum

FHSS



DSSS



IEEE 802.11 substandards

802.11a

- ◆ DSSS

802.11b

- ◆ OFDM

802.11g

- ◆ DSSS

802.11n

- ◆ a + b + g + MIMO

Bluetooth

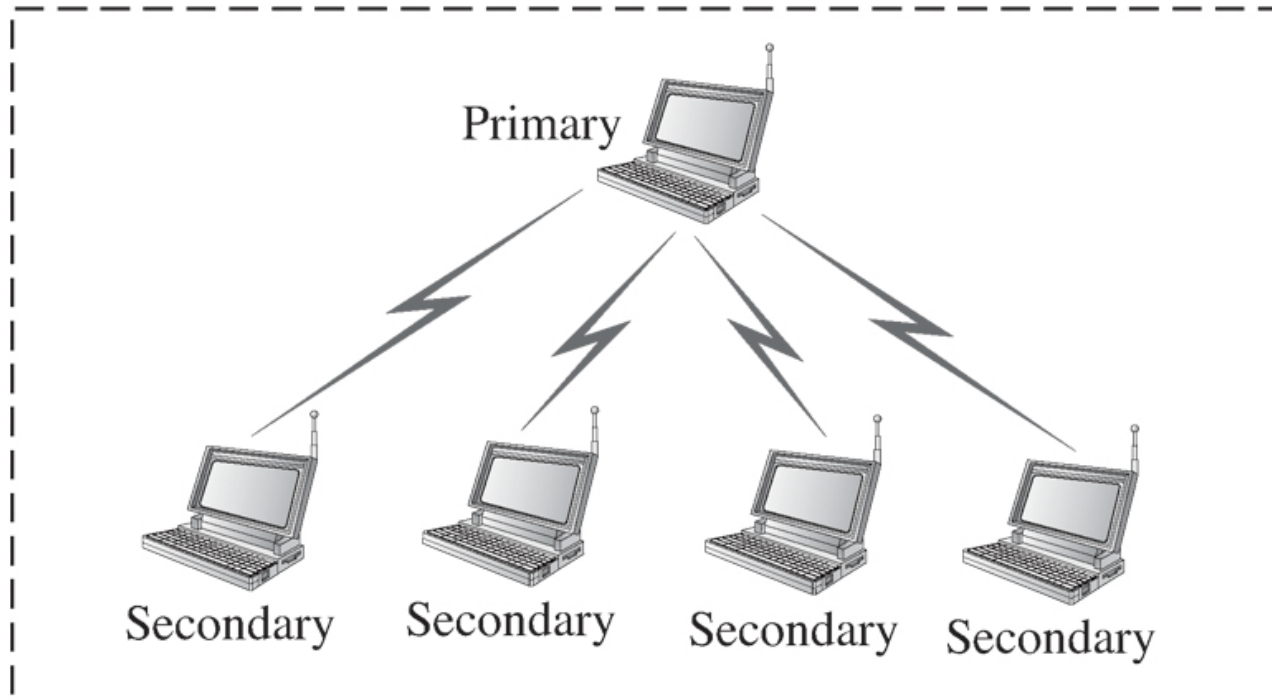
- ◆ Idea: Get rid of all the damn cables behind the computer
- ◆ Developed at LTH and Ericsson
- ◆ Implementation of IEEE 802.15

Bluetooth Piconet

- ◆ 1 master/primary
- ◆ Up to 7 slaves/secondary
- ◆ Up to 8 parked/resting slaves

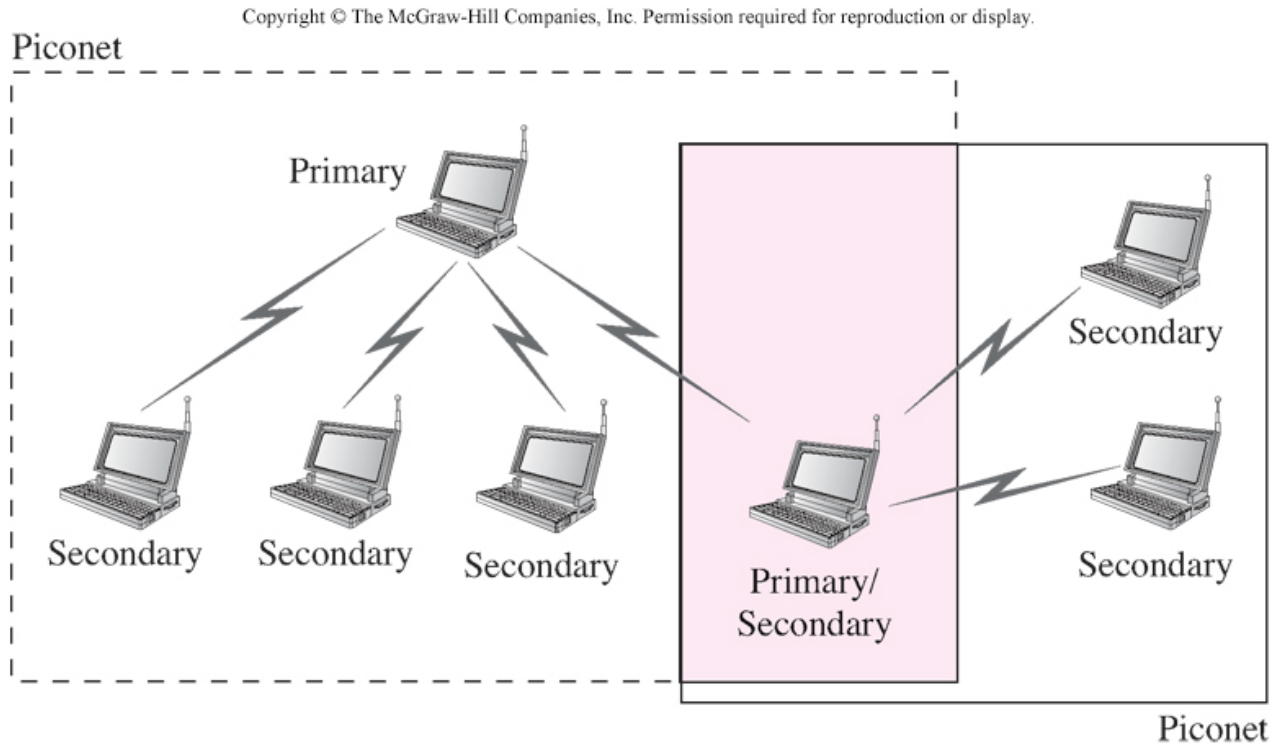
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Piconet



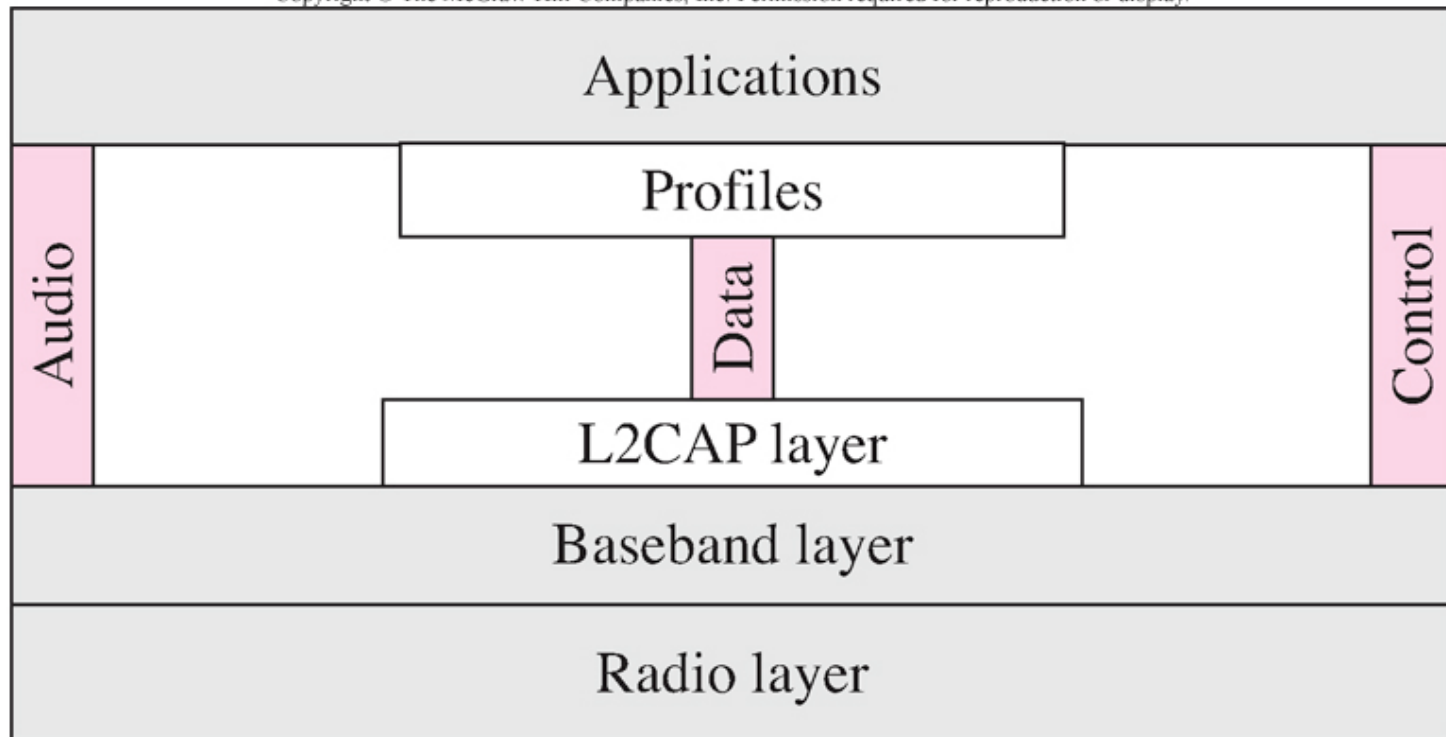
Bluetooth Scatternet

One master in one piconet is slave in another piconet



Bluetooth Layers

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Bluetooth Radio Layer

- ◆ 2,4 GHz (IMS band)
- ◆ FHSS
- ◆ 79 channels
- ◆ Each channel 1Mhz wide
- ◆ Changes channel 1600 times per second

Bluetooth Baseband Layer

- ◆ Similar to MAC layer in IEEE 802
- ◆ Access method: TDD-TDMA
 - Time Division Duplex
 - Time split half duplex

Datorkom 2006
Foreläs 5

